

$$\frac{u = 1}{f}$$

Power	.05	.10	.15	.20	.25	.30
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STATISTICAL POWER ANALYSIS for the BEHAVIORAL SCIENCES

$$\frac{u = 2}{f}$$

Power	.05	.10	.15	.20	.25	.30
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Second Edition

.50	475	119	53	30	20	14
.70	797	200	89	50	32	23
.80	1029	258	115	65	41	29
.90	1395	349	156	88	57	40
.95	1738	435	194	109	70	49
.99	4755	789	351	198	127	88

Jacob Cohen



$$\frac{u = 3}{f}$$

Power	.05	.10	.15	.20	.25	.30
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.50	419	105	47	27	18	12
.70	690	173	77	43	28	20
.80	883	221	99	56	36	25

**Statistical Power Analysis
for the Behavioral Sciences**
Second Edition

Statistical Power Analysis for the Behavioral Sciences

Second Edition

Jacob Cohen

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New York University
New York, New York*



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to Marcia and Aviva

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Preface to the Second Edition

In the quarter century that has passed since I first addressed power analysis (Cohen, 1962), and particularly during the decade that has elapsed since the revised edition of this book (1977), the escalation of the literature on power analysis has been difficult to keep up with.

In 1962, I published a survey of the articles in a volume of the *Journal of Abnormal and Social Psychology* from the perspective of their power to detect operationally defined small, medium, and large effect sizes [a meta-analysis before the term was coined (Bangert-Drowns, 1986)]. I found rather poor power, for example, a mean of .48 at the two-tailed .05 level for medium effect sizes.

Since the publication of the first edition (1969), there have been two or three dozen power surveys of either particular journals or topical areas, using its tables and (more or less) the same method. In addition to the half-dozen cited in the Preface to the Revised Edition in 1977, which were in the fields of counseling psychology, applied psychology, education, speech and hearing, and mass communication, there are numerous power surveys in many fields, for example: in educational research, in general education (Jones & Brewer, 1972), science education (Pennick & Brewer, 1972; Wooley & Dawson, 1983), English education (Daly & Hexamer, 1983), physical education (Christensen & Christensen, 1977), counselor education (Haase, 1974), social work education (Orme & Tolman, 1986) medical education (Wooley, 1983a), and educational measurement (Brewer & Owen, 1973). Power surveys have been done in social work and social intervention research (Crane, 1976; Judd & Kenny, 1981; Orme & Combs-Orme, 1986), in occupational therapy (Ottenbacher, 1982), abnormal psychology

(Sedlmeier & Gigerenzer, in press), personnel selection (Katzell & Dyer, 1977), and market research (Sawyer & Ball, 1981). A fairly large number have been accomplished in medicine: in clinical trials (Freiman, Chalmers, Smith, & Kuebler, 1977; Reed & Slaichert, 1981), public health (Wooley, 1983b), gerontology (Levenson, 1980), psychiatry (Rothpearl, Mohs, & Davis, 1981), and Australian medicine (Hall, 1982). Even further afield, a power survey was done in the field of geography (Bones, 1972). In addition to these published surveys, there have come to my attention about a dozen unpublished dissertations, research reports, and papers given at professional meetings surveying power in psychology, sociology, and criminology.

A corollary to the long neglect of power analysis is a relatively low awareness of the magnitude of phenomena in the behavioral sciences (Cohen, 1965). The emphasis on testing null hypotheses for statistical significance (R. A. Fisher's legacy) focused attention on the statistical significance of a result and away from the size of the effect being pursued (see Oakes, 1986; Gigerenzer, 1987; Chapter 11). A direct consequence of the recent attention to power, the last few years have witnessed a series of surveys of effect sizes: in social psychology (Cooper & Findlay, 1982), counseling psychology (Haase, Waechter, & Solomon, 1982), consumer behavior (Peterson, Albaum, & Beltramini, 1985), and market research (Sawyer & Ball, 1981).

The recent emergence of meta-analysis (Glass, McGaw, & Smith, 1981; Hedges & Olkin, 1985; Hunter, Schmidt, & Jackson, 1982; Kraemer, 1983) has been influenced by power analysis in the adoption of its effect size measures (Bangert-Drowns, 1986), and in turn, has had a most salutary influence on research progress and power analysis by revealing the level, variability, and correlates of the effect sizes operating in the areas to which it is applied.

The literature in power-analytic methodology has burgeoned during this period; pertinent references are given throughout this edition. Among the many topics here are applied power analysis for: nonstandard conditions (e.g., non-normality, heterogeneous variance, range restriction), non-parametric methods, various multiple comparison procedures, alternative methods of combining probabilities, and alternative stabilizing data transformations. There have been several articles offering simplified one-table methods of approximate power analysis including my own (1970) (which provided the basis for a chapter-length treatment in the Welkowitz, Ewen, & Cohen, 1982, introductory statistics text), Friedman (1982), and Kraemer (1985). The latter is particularly noteworthy in that it breaks new ground methodologically and is oriented toward teaching power analysis.

In marked contrast to the scene a decade or two ago, the current editions of the popular graduate level statistics textbooks oriented to the social and biological sciences provide at least some room for power analysis, and include working methods for the most common tests.

On the post-graduate front, as the word about power analysis has

spread, many “what is it” and “how to do it” articles have appeared in journals of widely diversified content, ranging from clinical pathology (Arkin, 1981) through applied psychology (Fagley, 1985) to biological community ecology (Toft & Shea, 1983).

Microcomputer programs for power analysis are provided by Anderson (1981), Dallal (1987), and Haase (1986). A program that both performs and teaches power analysis using Monte Carlo simulation is about to be published (Borenstein, M. & Cohen, J., 1988).

It would seem that power analysis has arrived.

Yet recently, two independent investigations have come to my attention that give me pause. Rossi, Rossi, and Cottril (in press), using the methods of my power survey of the articles in the 1960 volume of the *Journal of Abnormal and Social Psychology* (Cohen, 1962), performed power surveys of 142 articles in the 1982 volumes of the direct descendents of that journal, the *Journal of Personality and Social Psychology* and the *Journal of Abnormal Psychology*. When allowance is made for the slightly different (on the average) operational definitions of small, medium, and large effect sizes of the 1962 paper, there is hardly any change in power; for example, the mean power at the two-tailed .05 level for medium effect sizes of the 1982 articles was slightly above 50%, hardly different from the 48% in 1960.

Generally, the power surveys done since 1960 have found power not much better than I had. Some fields do show better power, but they are those in which subjects are easily come by, so the sample sizes used are larger than those in abnormal, personality, and social psychology: in educational research (Pennick & Brewer, 1972; Brewer & Owen, 1973), mass communication (Chase & Baran, 1976), applied psychology (Chase & Chase, 1975), and marketing research (Sawyer & Ball, 1981). However, there is no comparison of power over time in these areas.

Sedlmeier and Gigerenzer (in press) also studied the change in power since my 1962 results, using 54 articles in the 1984 volume of the *Journal of Abnormal Psychology*. They, too, found that the average power had not changed over the past 24-year period. In fact, when the power of the tests using experimentwise significance criteria (not encountered in my 1962 survey) were included, the median power for medium effects at the .05 level was .37. Even more dismaying is the fact that in seven articles, at least one of the null hypotheses was the research hypotheses, and the nonsignificance of the result was taken as confirmatory; the median power of these tests to detect a medium effect at the two-tailed .05 level was .25! In only two of the articles surveyed was power mentioned, and in none were there any power calculations. Sedlmeier and Gigerenzer's conclusion that my 1962 paper (and the extensive literature detailed above) “had no effect on actual practice” is consistent with the available evidence.

Yet, I find some solace from the following considerations: First, this may be a phenomenon on the abnormal-social-personality area and may not gen-

eralize to all behavioral-social-biological research areas. Second, to my certain knowledge, many journal editors and regular referees are quite knowledgeable about power and make editorial decisions in accordance with this knowledge. Third, I am told that some major funding entities require power analyses in grant applications. (I've even heard an unlikely story to the effect that in one of them there is a copy of this book in every office!) Finally, the research surveyed by Rossi et al. (in press) and Sedlmeier and Gigerenzer (in press), although published in the early 1980's, was mostly initiated in the late 1970's. The first edition of this book was not distributed until 1970. In the light of the fact that it took over three decades for Student's *t* test to come into general use by behavioral scientists, it is quite possible that there simply has not been enough time.

Taking all this into account, however, it is clear that power analysis has not had the impact on behavioral research that I (and other right-thinking methodologists) had expected. But we are convinced that it is just a matter of time.

This edition has the same approach and organization as its predecessors, but has some major changes from the Revised Edition.

1. A chapter has been added for power analysis in set correlation and multivariate methods (Chapter 10). Set correlation is a realization of the multivariate general linear model, and incorporates the standard multivariate methods (e.g., the multivariate analysis of variance and covariance) as special cases. While the standard methods are explicitly treated, the generality of set correlation offers a unifying framework and some new data-analytic possibilities (Cohen, 1982; Cohen & Cohen, 1983; Appendix 4).

2. A new chapter (Chapter 11) considers some general topics in power analysis in more integrated form than is possible in the earlier "working" chapters: effect size, psychometric reliability, and the efficacy of "qualifying" (differencing and partialling) dependent variables.

3. The two sets of working tables used for power and sample size determination in multiple regression and correlation analysis (Chapter 9) have been greatly expanded and provide more accurate values for a denser argument. These tables, derived from the noncentral *F* distribution, are also used for power and sample size determination in set correlation and multivariate methods (Chapter 10).

References have been updated and greatly expanded in keeping with the burgeoning increase in the literature of power analysis, and the errors in the previous edition, mostly caught by vigilant readers (to whom I offer my gratitude), corrected. I am surprised that I had to discover for myself the most egregious error of all: this edition does not presume, as did its predecessors, that all researchers are male.

As in the previous editions, I acknowledge the never ending learning pro-

cess afforded me by my students and consultees, and the continuing and unpayable debt of gratitude to my wife Patricia, who read, debated, and corrected all the new material despite a heavy workload of her own.

In their classic paper "Belief in the Law of Small Numbers," Tversky and Kahneman (1971) demonstrated how flawed are the statistical intuitions not only of psychologists in general, but even of mathematical psychologists. Most psychologists of whatever stripe believe that samples, even small samples, mirror the characteristics of their parent populations. In effect, they operate on the unstated premise that the law of large numbers holds for small numbers as well. They also believe that if a result is significant in one study, even if only barely so, it will most likely be significant in a replication, even if it has only half the sample size of the original. Tversky and Kahneman detail the various biases that flow from this "belief in the law of small numbers," and note that even if these biases cannot be easily unlearned, "the obvious precaution is computation. The believer in the law of small numbers has incorrect intuitions about significance level, power, and confidence intervals. Significance levels are usually computed and reported, but power and confidence limits are not. Perhaps they should be" (p. 110).

But as we have seen, too many of our colleagues have not responded to Tversky and Kahneman's admonition. It is almost as if they would rather follow W. H. Auden's proscription:

Thou shalt not sit
With statisticians nor commit
A social science.

They do so at their peril.

September, 1987

South Wellfleet, Massachusetts
Jacob Cohen

Preface to the Revised Edition

The structure, style, and level of this edition remain as in the original, but three important changes in content have been made:

1. Since the publication of the original edition, multiple regression/correlation analysis has been expanded into a very general and hence versatile system for data analysis, an approach which is uniquely suited to the needs of the behavioral sciences (Cohen and Cohen, 1975). A new chapter is devoted to an exposition of the major features of this data-analytic system and a detailed treatment of power analysis and sample size determination (Chapter 9).

2. The effect size index used for chi-square tests on frequencies and proportions (Chapter 7) has been changed from ϵ to $w(=\sqrt{\epsilon})$. This change was made in order to provide a more useful range of values and to make the operational definitions of "small," "medium," and "large" effect sizes for tests of contingency tables and goodness of fit consistent with those for other statistical tests (particularly those of Chapters 5 and 6). The formulas have been changed accordingly and the 84 look-up tables for power and sample size have been recomputed.

3. The original treatment of power analysis and sample size determination for the factorial design analysis of variance (Chapter 8) was approximate and faulty, yielding unacceptably large overestimation of power for main effects and underestimation for interactions. The treatment in this edition is materially changed and includes a redefinition of effect size for interactions.

The new method gives quite accurate results. Further insight into the analysis of variance is afforded when illustrative problems solved by the methods of this chapter are addressed and solved again by the multiple regression/correlation methods of the new Chapter 9.

Thus, this edition is substantially changed in the areas for which the original edition was most frequently consulted. In addition, here and there, some new material has been added (e.g., Section 1.5.5, "Proving" the Null Hypothesis) and some minor changes have been made for updating and correction.

In the seven years since the original edition was published, it has received considerable use as a supplementary textbook in intermediate level courses in applied statistics. It was most gratifying to note that, however slowly, it has begun to influence research planning and the content of textbooks in applied statistics. Several authors have used the book to perform power-analytic surveys of the research literature in different fields of behavioral science, among them Brewer (1972) in education (but see Cohen, 1973), Katzer and Sordt (1973) and Chase and Tucker (1975) in communication, Kroll and Chase (1975) in speech pathology, Chase and Baran (1976) in mass communication, and Chase and Chase (1976) in applied psychology; others are in preparation. Apart from their inherent value as methodological surveys, they have served to disseminate the ideas of power analysis to different audiences with salutary effects on them as both producers and consumers of research. It is still rare, however, to find power analysis in research planning presented in the introductory methods section of research reports (Cohen, 1973).

As in the original edition, I must first acknowledge my students and consultees, from whom I have learned so much, and then my favorite colleague, Patricia Cohen, a constant source of intellectual excitement and much more. I am grateful to Patra Lindstrom for the exemplary fashion in which she performed the exacting chore of typing the new tables and manuscript.

NEW YORK
JUNE 1976

JACOB COHEN

Preface to the Original Edition

During my first dozen years of teaching and consulting on applied statistics with behavioral scientists, I became increasingly impressed with the importance of statistical power analysis, an importance which was increased an order of magnitude by its neglect in our textbooks and curricula. The case for its importance is easily made: What behavioral scientist would view with equanimity the question of the probability that his investigation would lead to statistically significant results, i.e., its power? And it was clear to me that most behavioral scientists not only could not answer this and related questions, but were even unaware that such questions were answerable. Casual observation suggested this deficit in training, and a review of a volume of the *Journal of Abnormal and Social Psychology* (JASP) (Cohen, 1962), supported by a small grant from the National Institute of Mental Health (M-5174A), demonstrated the neglect of power issues and suggested its seriousness.

The reason for this neglect in the applied statistics textbooks became quickly apparent when I began the JASP review. The necessary materials for power analysis were quite inaccessible, in two senses: they were scattered over the periodical and hardcover literature, and, more important, their use assumed a degree of mathematical sophistication well beyond that of most behavioral scientists.

For the purpose of the review, I prepared some sketchy power look-up tables, which proved to be very easily used by the students in my courses at New York University and by my research consultees. This generated the

idea for this book. A five-year NIMH grant provided the support for the program of research, system building, computation, and writing of which the present volume is the chief product.

The primary audience for which this book is intended is the behavioral or biosocial scientist who uses statistical inference. The terms "behavioral" and "biosocial" science have no sharply defined reference, but are here intended in the widest sense and to include the academic sciences of psychology, sociology, branches of biology, political science and anthropology, economics, and also various "applied" research fields: clinical psychology and psychiatry, industrial psychology, education, social and welfare work, and market, political polling, and advertising research. The illustrative problems, which make up a large portion of this book, have been drawn from behavioral or biosocial science, so defined.

Since statistical inference is a logical-mathematical discipline whose applications are not restricted to behavioral science, this book will also be useful in other fields of application, e.g., agronomy and industrial engineering.

The amount of statistical background assumed in the reader is quite modest: one or two semesters of applied statistics. Indeed, all that I really assume is that the reader knows how to proceed to perform a test of statistical significance. Thus, the level of treatment is quite elementary, a fact which has occasioned some criticism from my colleagues. I have learned repeatedly, however, that the *typical* behavioral scientist approaches applied statistics with considerable uncertainty (if not actual nervousness), and requires a verbal-intuitive exposition, rich in redundancy and with many concrete illustrations. This I have sought to supply. Another feature of the present treatment which should prove welcome to the reader is the minimization of required computation. The extensiveness of the tables is a direct consequence of the fact that most uses will require no computation at all, the necessary answers being obtained directly by looking up the appropriate table.

The sophisticated applied statistician will find the exposition unnecessarily prolix and the examples repetitious. He will, however, find the tables useful. He may also find interesting the systematic treatment of population effect size, and particularly the proposed conventions or operational definitions of "small," "medium," and "large" effect sizes defined across all the statistical tests. Whatever originality this work contains falls primarily in this area.

This book is designed primarily as a handbook. When so used, the reader is advised to read Chapter 1 and then the chapter which treats the specific statistical test in which he is interested. I also suggest that he read all the relevant illustrative examples, since they are frequently used to carry along the general exposition.

The book may also be used as a supplementary textbook in intermediate level courses in applied statistics in behavioral/biosocial science. I have been

using it in this way. With relatively little guidance, students at this level quickly learn both the concepts and the use of the tables. I assign the first chapter early in the semester and the others in tandem with their regular textbook's treatment of the various statistical tests. Thus, each statistical test or research design is presented in close conjunction with power-analytic considerations. This has proved most salutary, particularly in the attention which must then be given to anticipated population effect sizes.

Pride of place, in acknowledgment, must go to my students and consultees, from whom I have learned much. I am most grateful to the memory of the late Gordon Ierardi, without whose encouragement this work would not have been undertaken. Patricia Waly and Jack Huber read and constructively criticized portions of the manuscript. I owe an unpayable debt of gratitude to Joseph L. Fleiss for a thorough technical critique. Since I did not follow all his advice, the remaining errors can safely be assumed to be mine. I cannot sufficiently thank Catherine Henderson, who typed much of the text and all the tables, and Martha Plimpton, who typed the rest.

As already noted, the program which culminated in this book was supported by the National Institute of Mental Health of the Public Health Service under grant number MH-06137, which is duly acknowledged. I am also most indebted to Abacus Associates, a subsidiary of American Bioculture, Inc., for a most generous programming and computing grant which I could draw upon freely.

NEW YORK
JUNE 1969

JACOB COHEN

The Concepts of Power Analysis

The power of a statistical test is the probability that it will yield statistically significant results. Since statistical significance is so earnestly sought and devoutly wished for by behavioral scientists, one would think that the *a priori* probability of its accomplishment would be routinely determined and well understood. Quite surprisingly, this is not the case. Instead, if we take as evidence the research literature, we find evidence that statistical power is frequently not understood and, in reports of research where it is clearly relevant, the issue is not addressed.

The purpose of this book is to provide a self-contained comprehensive treatment of statistical power analysis from an “applied” viewpoint. The purpose of this chapter is to present the basic conceptual framework of statistical hypothesis testing, giving emphasis to power, followed by the framework within which this book is organized.

1.1 GENERAL INTRODUCTION

When the behavioral scientist has occasion to don the mantle of the applied statistician, the probability is high that it will be for the purpose of testing one or more null hypotheses, i.e., “the hypothesis that the phenomenon to be demonstrated is in fact absent [Fisher, 1949, p. 13].” Not that he hopes to “prove” this hypothesis. On the contrary, he typically hopes to “reject” this hypothesis and thus “prove” that the phenomenon in question is in fact present.

Let us acknowledge at the outset the necessarily probabilistic character of statistical inference, and dispense with the mocking quotation marks

about words like *reject* and *prove*. This may be done by requiring that an investigator set certain appropriate probability standards for research results which provide a basis for rejection of the null hypothesis and hence for the proof of the existence of the phenomenon under test. Results from a random sample drawn from a population will only approximate the characteristics of the population. Therefore, even if the null hypothesis is, in fact, true, a given sample result is not expected to mirror this fact exactly. Before sample data are gathered, therefore, the investigator selects some prudently small value α (say .01 or .05), so that he *may* eventually be able to say about his sample data, "*If the null hypothesis is true, the probability of the obtained sample result is no more than α ,*" i.e. a statistically significant result. *If* he can make this statement, since α is small, he said to have rejected the null hypothesis "with an α significance criterion" or "at the α significance level." If, on the other hand, he finds the probability to be greater than α , he cannot make the above statement and he has failed to reject the null hypothesis, or, equivalently finds it "tenable," or "accepts" it, all at the α significance level. Note that α is set in advance.

We have thus isolated one element of this form of statistical inference, the standard of proof that the phenomenon exists, or, equivalently, the standard of disproof of the null hypothesis that states that the phenomenon does not exist.

Another component of the significance criterion concerns the exact definition of the nature of the phenomenon's existence. This depends on the details of how the phenomenon is manifested and statistically tested, e.g., the directionality/nondirectionality ("one tailed"/"two tailed") of the statement of the alternative to the null hypothesis.¹ When, for example, the investigator is working in a context of comparing some parameter (e.g., mean, proportion, correlation coefficient) for two populations A and B, he can define the existence of the phenomenon in two different ways:

1. The phenomenon is taken to exist if the parameters of A and B differ. No direction of the difference, such as A larger than B, is specified, so that departures in either direction from the null hypothesis constitute evidence against it. Because either tail of the sampling distribution of differences may contribute to α , this is usually called a two-tailed or two-sided test.

2. The phenomenon is taken to exist only if the parameters of A and B differ in a direction specified in advance, e.g., A larger than B. In this

¹ Some statistical tests, particularly those involving comparisons of more than two populations, are naturally nondirectional. In what immediately follows, we consider those tests which contrast two populations, wherein the experimenter ordinarily explicitly chooses between a directional and nondirectional statement of his alternate hypothesis. See below, Chapters 7 and 8.

circumstance, departures from the null hypothesis only in the direction specified constitute evidence against it. Because only one tail of the sampling distribution of differences may contribute to α , this is usually called a one-tailed or one-sided test.

It is convenient to conceive of the significance criterion as embodying both the probability of falsely rejecting the null hypothesis, α , and the "sidedness" of the definition of the existence of the phenomenon (when relevant). Thus, the significance criterion on a two-tailed test of the null hypothesis at the .05 significance level, which will be symbolized as $\alpha_2 = .05$, says two things: (a) that the phenomenon whose existence is at issue is understood to be manifested by any difference between the two populations' parameter values, and (b) that the standard of proof is a sample result that would occur less than 5% of the time if the null hypothesis is true. Similarly, a prior specification defining the phenomenon under study as that for which the parameter value for A is larger than that of B (i.e., one-tailed) and the probability of falsely rejecting the null is set at .10 would be symbolized as a significance criterion of $\alpha_1 = .10$. The combination of the probability and the sidedness of the test into a single entity, the significance criterion, is convenient because this combination defines in advance the "critical region," i.e., the range of values of the outcome which leads to rejection of the null hypothesis and, perforce, the range of values which leads to its nonrejection. Thus, when an investigator plans a statistical test at some given significance criterion, say $\alpha_1 = .10$, he has effected a specific division of all the possible results of his study into those which will lead him to conclude that the phenomenon exists (with risk α no greater than .10 and a one-sided definition of the phenomenon) and those which will not make possible that conclusion.²

The above review of the logic of classical statistical inference reduces to a null hypothesis and a significance criterion which defines the circumstances which will lead to its rejection or nonrejection. Observe that the significance criterion embodies the risk of mistakenly rejecting a null hypothesis. The entire discussion above is conditional on the truth of the null hypothesis.

But what if, indeed, the phenomenon *does* exist and the null hypothesis is *false*? This is the usual expectation of the investigator, who has stated the null hypothesis for tactical purposes so that he may reject it and conclude that the phenomenon exists. But, of course, the fact that the phenomenon exists in the population far from guarantees a statistically significant result,

² The author has elsewhere expressed serious reservations about the use of directional tests in psychological research in all but relatively limited circumstances (Cohen, 1965). The bases for these reservations would extend to other regions of behavioral science. These tests are however of undoubted statistical validity and in common use, so he has made full provision for them in this work.

i.e., one which warrants the conclusion that it exists, for this conclusion depends upon meeting the agreed-upon standard of proof (i.e., significance criterion). It is at this point that the concept of statistical power must be considered.

The power of a statistical test of a null hypothesis is the probability that it will lead to the rejection of the null hypothesis, i.e., the probability that it will result in the conclusion that the phenomenon exists. Given the characteristics of a specific statistical test of the null hypothesis and the state of affairs in the population, the power of the test can be determined. It clearly represents a vital piece of information about a statistical test applied to research data (cf. Cohen, 1962). For example, the discovery, during the planning phase of an investigation, that the power of the eventual statistical test is low should lead to a revision in the plans. As another example, consider a completed experiment which led to nonrejection of the null hypothesis. An analysis which finds that the power was low should lead one to regard the negative results as ambiguous, since failure to reject the null hypothesis cannot have much substantive meaning when, even though the phenomenon exists (to some given degree), the *a priori* probability of rejecting the null hypothesis was low. A detailed consideration of the use of power analysis in planning investigations and assessing completed investigations is reserved for later sections.

The power of a statistical test depends upon three parameters: the significance criterion, the reliability of the sample results, and the "effect size," that is, the *degree* to which the phenomenon exists.

1.2 SIGNIFICANCE CRITERION

The role of this parameter in testing null hypotheses has already been given some consideration. As noted above, the significance criterion represents the standard of proof that the phenomenon exists, or the risk of mistakenly rejecting the null hypothesis. As used here, it directly implies the "critical region of rejection" of the null hypothesis, since it embodies both the probability of a class of results given that the null hypothesis is true (α), as well as the definition of the phenomenon's existence with regard to directionality. For power to be defined, its value must be set in advance.

The significance level, α , has been variously called the error of the first kind, the Type I error, and the alpha error. Since it is the rate of rejecting a true null hypothesis, it is taken as a relatively small value. It follows then that the smaller the value, the more rigorous the standard of null hypothesis rejection or, equivalently, of proof of the phenomenon's existence. Assume that a phenomenon exists in the population to some given degree. Other things equal, the more stringent the standard for proof, i.e., the lower the value of α , the poorer the chances are that the sample will provide results

which meet this standard, i.e., the lower the power. Concretely, if an investigator is prepared to run only a 1% risk of false rejection of the null hypothesis, the probability of his data meeting this standard is lower than would be the case were he prepared to use the less stringent standard of a 10% risk of false rejection.

The practice of taking α very small ("the smaller the better") then results in power values being relatively small. However, the complement of the power ($1 - \text{power}$), here symbolized as β , is also error, called Type II or beta error, since it represents the "error" rate of failing to reject a false null hypothesis. Thus it is seen that statistical inference can be viewed as weighing, in a manner relevant to the substantive issues of an investigation, these two kinds of errors. An investigator can set the risk of false null hypothesis rejection at a vanishingly small level, say $\alpha = .001$, but in so doing, he may reduce the power of his test to .10 (hence beta error probability, β , is $1 - .10 = .90$). Two comments may be made here:

1. The general neglect of issues of statistical power in behavioral science may well result, in such instances, in the investigator's failing to realize that the $\alpha = .001$ value leads in his situation to $\text{power} = .10$, $\beta = .90$ (Cohen, 1962). Presumably, although not necessarily, such a realization would lead to a revision of experimental plans, including possibly an upward revision of the α level to increase power.

2. If the investigator proceeds as originally planned, he implies a conception of the relative seriousness of Type I to Type II error (risk of false null rejection to risk of false null acceptance) of $\beta/\alpha = .90/.001 = 900$ to 1, i.e., he implicitly believes that mistakenly rejecting the null hypothesis under the assumed conditions is 900 times more serious than mistakenly accepting it. In another situation, with $\alpha = .05$, $\text{power} = .80$, and hence $\beta = 1 - .80 = .20$, the relative seriousness of Type I to Type II error is $\beta/\alpha = .20/.05 = 4$ to 1; thus mistaken rejection of the null hypothesis is considered four times as serious as mistaken acceptance.

The directionality of the significance criterion (left unspecified in the above examples) also bears on the power of a statistical test. When the null hypothesis can be rejected in *either* direction so that the critical significance region is in *both* tails of the sampling distribution of the test statistic (e.g., a t ratio), the resulting test will have less power than a test at the same α level which is directional, *provided that* the sample result is in the direction predicted. Since directional tests cannot, by definition, lead to rejecting the null hypothesis in the direction *opposite* to that predicted, these tests have no power to detect such effects. When the experimental results are in the predicted direction, all other things equal, a test at level α_1 will have power equal for all practical purposes to a test at $2\alpha_2$.

Concretely, if an experiment is performed to detect a difference between the means of populations A and B, say m_A and m_B , in *either* direction at the $\alpha_2 = .05$ significance criterion, under given conditions, the test will have a certain power. If, instead, an anticipation of m_A greater than m_B leads to a test at $\alpha_1 = .05$, this test will have power approximately equal to a two-tailed test with $\alpha_2 = .10$, hence greater power than the test at $\alpha_2 = .05$, provided that in fact m_A is greater than m_B . If m_B is greater than m_A , the test at $\alpha_1 = .05$ has *no* power, since that conclusion is inadmissible. The temptation to perform directional tests because of their greater power at the same α level should be tempered by the realization that they preclude finding results opposite to those anticipated. There are occasional circumstances where the nature of the decision is such that the investigator does not need to know about effects in the opposite direction. For example, he will take a certain course of action if m_A is greater than m_B and not otherwise. If otherwise, he does not need to distinguish between their equality and m_B greater than m_A . In such infrequent instances, one-tailed tests are appropriate (Cohen, 1965, pp. 106–111).

In the tables in this book, provision is made for tests at the .01, .05, and .10 significance levels. Where a statistical test may ordinarily be performed either nondirectionally or directionally, both α_2 and α_1 tables are provided. Since power for $\alpha_1 = .05$ is virtually identical with power for $\alpha_2 = .10$, a single power table suffices. Similarly, tables for $\alpha_1 = .01$ provide values for $\alpha_2 = .02$, and tables for $\alpha_1 = .10$ values for $\alpha_2 = .20$; also, tables for $\alpha_2 = .01$ provide values for $\alpha_1 = .005$, tables at $\alpha_2 = .05$ provide values for $\alpha_1 = .025$.

1.3 RELIABILITY OF SAMPLE RESULTS AND SAMPLE SIZE

The reliability (or precision) of a sample value is the closeness with which it can be expected to approximate the relevant population value. It is necessarily an estimated value in practice, since the population value is generally unknown. Depending upon the statistic in question, and the specific statistical model on which the test is based, reliability may or may not be directly dependent upon the unit of measurement, the population value, and the shape of the population distribution. However, it is *always* dependent upon the size of the sample.

For example, one conventional means for assessing the reliability of a statistic is the standard error (SE) of the statistic. If we consider the arithmetic mean of a variable X ($\bar{X}$), its reliability may be estimated by the standard error of the mean,

$$SE_{\bar{X}} = \sqrt{\frac{s^2}{n}},$$

where s^2 is the usual unbiased estimate (from the random sample) of the

population variance of $\mathbf{X}$, and n is the number of independent units in (i.e., the size of) the sample.

Concretely, if a sample of $n = 49$ cases yields a variance estimate for IQ of 196, then the standard error of the mean is given by

$$SE_{\bar{X}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{196}{49}} = 2.$$

Thus, sample means based on 49 cases can be expected to have variability as measured by their own standard deviation of 2 IQ units. Clearly the greater the degree to which means of different samples vary among themselves, the less any of them can be relied upon, i.e., the less the reliability of the mean of the sample in hand. Note that in this instance reliability depends upon the unit of measurement (IQ) and sample size, but not on the value of the population mean or (to any material degree) on the shape of the IQ distribution.

On the other hand, consider the sampling reliability of a product moment coefficient of correlation, r . Its standard error is

$$SE_r = \frac{1 - r_p^2}{\sqrt{n - 1}},$$

where

r_p = the population value of r , and

n = the number of paired observations in the sample.

Note that the reliability of the sample r depends upon the magnitude of the (generally unknown) population r_p value and n , but not on the units in which the correlated variables are measured.

Not all statistical tests involve the explicit definition of a standard error of a sample value, but all do involve the more general conception of sample reliability. Moreover, and most important, whatever else sample reliability may be dependent upon, it *always* depends upon the size of the sample.

The nature of the dependence of reliability upon n is obvious from the illustrative formulas, and, indeed, intuitively. The larger the sample size, other things being equal, the smaller the error and the greater the reliability or precision of the results. The further relationship with power is also intuitively evident: the greater the precision of the sample results, other things being equal, the greater the probability of detecting a nonnull state of affairs, i.e., the more clearly the phenomenon under test can manifest itself against the background of (experimentally irrelevant) variability. Thus, we can directly formulate the relationship between sample size and power. As is intuitively obvious, increases in sample size increase statistical power, the probability of detecting the phenomenon under test.

Focusing on sample size as an invariant factor in power should not make

the researcher lose sight of the fact that other research elements potentially under his control also affect power. Random measurement error, be it due to psychometric unreliability, observational carelessness, dirty testtubes, or any other source, because it increases the variability of the observations beyond their necessary "true" variability, also reduces the precision of sample results and thus reduces power. In general, *anything* which reduces the variability of observations by the exclusion of sources of variability which are irrelevant to the assessment of the phenomenon under study will serve to increase power. Experimental design is an area of inquiry wholly devoted to the removal of irrelevant sources of variability for the increase of precision and therefore for the increase of the statistical power of tests of null hypotheses (cf. Cox, 1958).

In this book, provision is made for the accomplishment of power analyses for the statistical tests associated with the most frequently utilized experimental designs and their accompanying null hypotheses. Issues such as the effects of a given level of random measurement error on power are not explicitly provided for. Sample size, the invariant feature of sample precision, is, however, a factor in all the power tables. It is used in both of the major kinds of analysis tables herein provided; in the power tables, sample size is one of the elements used to determine the power of the test, and in the sample size tables, it is the dependent variable of the function of the desired level of power (in both instances under given conditions of significance criterion and population effect size).

1.4 THE EFFECT SIZE

To this point, the phenomenon in the population under statistical test was considered as either absent (null hypothesis true) or present (null hypothesis false). The absence of the phenomenon implies some specific value for a population parameter. For example, in a study to determine whether there is a sex difference in incidence of paranoid schizophrenia, the investigator may draw a sample of patients bearing that diagnosis from the relevant population and determine the proportion of males. The null hypothesis being tested is that the population proportion of males is .50, a specific value.^{3,4} Equivalently, we might say that the size of the "effect" of sex on the presence of

³ The assumption is made here that .50 is the proportion of males in the population of interest.

⁴ For the sake of simplicity, the null hypothesis is treated in this section for the non-directional form of the significance criterion. For example, a directional (one-tailed) test here that the male proportion is greater than .50 implies a null hypothesis that it is equal to or less than .50. The reader may supply his own necessary qualifications of the null hypothesis for the directional case in each illustration.

the diagnosis is zero. In another study concerned with the IQs of children born in multiple births, the null hypothesis might be that the multiple birth population in question has a mean IQ of 100 (i.e., the general population mean), again a specific value, or that the size of effect of being part of a multiple birth on IQ is zero. As yet another example of a one-sample test, in a study of the construct validity of a neurophysiological measure of introversion-extroversion, its product moment r with an accepted questionnaire measure for a sample of college students is determined. The null hypothesis here is that the population r is zero, or that the effect size of either on the other is zero.

In circumstances where two populations are being compared, the null hypothesis usually takes the form "the difference in the value of the relevant parameters is zero," a specific value. Thus, in a consumer survey research to determine whether preference for a particular brand A over its chief competitor B is related to the income level of the consumer, the null hypothesis might be: The difference in median family income of brand A and brand B users is zero, or, equivalently, that the size of the effect of income on brand preference is zero. Or, in a personnel selection study to determine which of two screening tests, A or B, is a better predictor of performance ratings (C), the null hypothesis might take the form: The difference between population product moment r 's of A with C and B with C is zero.

Statistical tests involving more than two samples test null hypotheses that imply the constancy of a parameter over the populations involved. The literal statement of the null hypothesis depends upon the specific test involved. For example, the F test of the analysis of variance for $k \geq 2$ means has as its null hypothesis the proposition that the variance of a set of population means is zero, a condition that can only obtain when they are equal. Similarly, a test of whether a set of $k \geq 2$ population proportions are equal can be performed by means of the chi-square statistic. The null hypothesis here is that the variance of the population proportions equals zero (an exact value), a condition which can only obtain when they are all equal. In both of these instances we can think of the null hypothesis as the circumstance in which differences in the independent variable, the k populations, have no effect (have an effect size of zero) on the means or proportions of the dependent variable.

Thus, we see that the absence of the phenomenon under study is expressed by a null hypothesis which specifies an exact value for a population parameter, one which is appropriate to the way the phenomenon under study is manifested. Without intending any necessary implication of causality, it is convenient to use the phrase "effect size" to mean "the *degree* to which the phenomenon is present in the population," or "the degree to which the

null hypothesis is false." Whatever the manner of representation of a phenomenon in a particular research in the present treatment, the null hypothesis always means that the effect size is zero.

By the above route, it can now readily be made clear that when the null hypothesis is false, it is false to some specific degree, i.e., *the effect size (ES) is some specific nonzero value in the population*. The larger this value, the greater the *degree* to which the phenomenon under study is manifested. Thus, in terms of the previous illustrations:

1. If the percentage of males in the population of psychiatric patients bearing a diagnosis of paranoid schizophrenia is 52%, and the effect is measured as a departure from the hypothesized 50%, the ES is 2%; if it is 60%, the ES is 10%, a larger ES.
2. If children of multiple births have a population mean IQ of 96, the ES is 4 IQ units (or -4 , depending on directionality of significance criterion); if it is 92, the ES is 8 (or -8) IQ units, i.e., a larger ES.
3. If the population product moment r between neurophysiological and questionnaire measures of introversion-extroversion is .30, the ES is .30; if the r is .60, so is the ES, a larger value and a larger departure from the null hypothesis, which here is $r = 0$.
4. If the population of consumers preferring brand A has a median annual income \$700 higher than that of brand B, the ES is \$700. If the population median difference and hence the ES is \$1000, the effect of income on brand preference would be larger.

Thus, whether measured in one unit or another, whether expressed as a difference between two population parameters or the departure of a population parameter from a constant or in any other suitable way, the ES can itself be treated as a parameter which takes the value zero when the null hypothesis is true and *some other specific nonzero value* when the null hypothesis is false, and in this way the ES serves as an index of degree of departure from the null hypothesis.

The reasons that the above discussion has proceeded in such redundant detail are twofold. On the one hand, ES is in practice a most important determinant of power or required sample size or both, and on the other hand, it is the least familiar of the concepts surrounding statistical inference among practicing behavior scientists. The reason for the latter, in turn, can be found in the difference in null hypothesis testing between the procedures of Fisher (1949) and those of Neyman and Pearson (1928, 1933).

The Fisherian formulation posits the null hypothesis as described above, i.e., the ES is zero, to which the "alternative" hypothesis is that the ES is *not* zero, i.e., *any* nonzero value. Without further specification, although null hypotheses may be tested and thereupon either rejected or not rejected,

no basis for statistical power analysis exists. By contrast, the Neyman-Pearson formulation posits an *exact* alternative for the ES, i.e., the *exact* size of the effect the experiment is designed to detect. With an exact alternative hypothesis or specific nonzero ES to be detected, given the other elements in statistical inference, statistical power analysis may proceed.

Thus, in the previous illustrations, the statements about possible population ES values (e.g., “if the population product moment r between neurophysiological and questionnaire measures of introversion-extroversion is .30, the ES is .30”) are statements of alternative hypotheses.

The relationship between ES and power should also be intuitively evident. The larger the ES posited, other things (significance criterion, sample size) being equal, the greater the power of the test. Similarly, the relationship between ES and necessary sample size: the larger the ES posited, other things (significance criterion, desired power) being equal, the smaller the sample size necessary to detect it.

To this point, the ES has been considered quite abstractly as a parameter which can take on varying values (including zero in the null case). In any given statistical test, it must be indexed or measured in some defined unit appropriate to the data, test, and statistical model employed. In the previous illustrations, ES was variously expressed as a departure in percent from 50, a departure in IQ units from 100, a product moment r , a difference between two medians in dollars, etc. It is clearly desirable to reduce this diversity of units as far as possible, consistent with present usage by behavioural scientists. From one point of view, a universal ES index, applicable to all the various research issues and statistical models used in their appraisal, would be the ideal. Apart from some formidable mathematical-statistical problems in the way, even if such an ideal could be achieved, the result would express ES in terms so unfamiliar to the researcher in behavioral science as to be self-defeating.

However, some generalization is obviously necessary. One cannot prepare a set of power tables for each new measurement unit with which one works. That is, the researcher who plans a test for a difference in mean IQs must use the same power tables as another who plans a test for a difference in mean weights, just as they will use the same tables of t when the research is performed. t is a “pure” (dimensionless) number, one free of raw unit, as are also, for example, correlation coefficients or proportions of variance. Thus, as will be seen in Chapter 2, the ES index for differences between population means is standardized by division by the common within-population standard deviation (σ), i.e., the ES here is not the difference between mean “raw” scores, but the difference between mean “ z ” standard scores (Hays, 1981), or the mean difference expressed in within-population σ units. In the F test for $k \geq 2$ population means, the ES also uses such standardized means;

in testing "main effects" in the analysis of variance the ES is *their* standard deviation, σ_m , the standard deviation of standardized means (Chapter 8).

Each test for which power tables are provided thus has a metric-free ES index appropriate to it. A higher order of generalization is frequently possible. Specifically, several ES indices can be translated into the proportion of variance (PV) accounted for in the dependent variable. Where this is possible, it is discussed in the introductory material for the test. Also, each ES index chosen usually relates to yet other commonly used indices and these are also described in the same place.

The behavior scientist who comes to statistical power analysis may find himself grappling with the problem of what ES to posit as an alternate to the null hypothesis, or, more simply, how to answer the questions "How large an effect do I expect exists in the population?" He may initially find it difficult to answer the question even in general terms, i.e., "small" or "large," let alone in terms of the specific ES index demanded. Being forced to think in more exact terms than demanded by the Fisherian alternative (ES is any nonzero value) is likely to prove salutary. He can call upon theory for some help in answering the question and on his critical assessment of prior research in the area for further help. When these are supplemented with the understanding of the ES index provided in the introductory material to the relevant chapter, he can decide upon the ES value to adopt as an alternative to the null.

When the above has not provided sufficient guidance, the reader has an additional recourse. For each statistical test's ES index, the author proposes, *as a convention*, ES values to serve as operational definitions of the qualitative adjectives "small," "medium," and "large." This is an operation fraught with many dangers: The definitions are arbitrary, such qualitative concepts as "large" are sometimes understood as absolute, sometimes as relative; and thus they run a risk of being misunderstood.

In justification, several arguments may be offered. It must first be said that all conventions are arbitrary. One can only demand of them that they not be unreasonable. Also, all conventions may be misused and their conventional status thus abused. For example, the .05 significance criterion, although unofficial, has come to serve as a convention for a (minimum) basis for rejecting the null hypothesis in most areas of behavioral and biological science. Unfortunately, its status as only a convention is frequently ignored; there are many published instances where a researcher, in an effort at rectitude, fails to report that a much desired null rejection would be possible at the .06 level but instead treats the problem no differently than he would have had it been at the .50 level! Still, it is convenient that "significance" without further specification can be taken to mean "significance at no more than the .05 level."

Although arbitrary, the proposed conventions will be found to be reasonable by reasonable people. An effort was made in selecting these operational criteria to use levels of ES which accord with a subjective average of effect sizes such as are encountered in behavioral science. "Small" effect sizes must not be so small that seeking them amidst the inevitable operation of measurement and experimental bias and lack of fidelity is a bootless task, yet not so large as to make them fairly perceptible to the naked observational eye. Many effects sought in personality, social, and clinical-psychological research are likely to be small effects as here defined, both because of the attenuation in validity of the measures employed and the subtlety of the issues frequently involved. In contrast, large effects must not be defined as so large that their quest by statistical methods is wholly a labor of supererogation, or to use Tukey's delightful term "statistical sanctification." That is, the difference in size between apples and pineapples is of an order which hardly requires an approach via statistical analysis. On the other side, it cannot be defined so as to encroach on a reasonable range of values called medium. Large effects are frequently at issue in such fields as sociology, economics, and experimental and physiological psychology, fields characterized by the study of potent variables or the presence of good experimental control or both.

Since effects are appraised against a background of random variation, the control of various sources of variation through the use of improved research designs serves to increase effect sizes as they are defined here. A simple example of this is a study of sex difference in some defined ability. Assume that a difference of 4 score points exists between male and female population means, where each population has a standard deviation of 16. A research plan which randomly samples the two populations (simple randomized design or comparison between two independent means) is operating with an ES of $4/16 = .25$. Another research plan might proceed by comparing means of males and their sisters (comparison of two dependent means). Now, these populations can also be assumed to have a mean difference of 4 score points, but because of the removal of the variation between families afforded by this design (or equivalently when allowance is made for the brother-sister correlation in the ability), the *effective* standard deviation will be reduced to the fraction $\sqrt{1-r}$ of 16, say to 12 (when r between siblings = .44), and the actual ES operating in the situation is $4/12 = .33$, a larger value than for the simple randomized design. Thus, *operative* effect sizes may be increased not only by improvement in measurement and experimental technique, but also by improved experimental designs.

Each of the Chapters 2-10 will present in some detail the ES index appropriate to the test to which the chapter is devoted. Each will be translated into alternative forms, the operational definitions of "small," "medium," and "large" will be presented, and examples drawn from various fields will

illustrate the test. This should serve to clarify the ES index involved and make the methods and tables useful in research planning and appraisal. Finally, in Chapter 11, Section 11.1 is devoted to a general consideration of ES in the behavioral sciences.

1.5 TYPES OF POWER ANALYSIS

Four parameters of statistical inference have been described: power, significance criterion (α), sample size (n), and effect size (ES). They are so related that any one of them is a function of the other three, which means that when any three of them are fixed, the fourth is completely determined. This relationship makes formally possible four types of power analysis; in each, one of these parameters is determined as a function of the other three (Cohen, 1965, pp. 97-101).

1.5.1 POWER AS A FUNCTION OF α , ES, AND n . The preceding material has been largely oriented toward the type of analysis in which, given the specification of α , ES, and n , power is determined. For example, an investigator plans a test of the significance of a product moment r at $\alpha_2 = .05$ using $n = 30$ cases. The ES he wishes to detect is a population r of .40. Given these specifications, he finds (by the methods of Section 3.3 in Chapter 3) that power equals .61. He may then decide to change his specifications to increase power.

Such analyses are usefully performed as part of research planning. They can also be performed on completed studies to determine the power which a given statistical test had, as in the power survey of the studies in a volume of the *Journal of Abnormal and Social Psychology* (Cohen, 1962). In each of Chapters 2-10, the power tables (numbered B.3.A., where B is the chapter number and A indexes the significance criterion) are designed for this type of analysis. The sections designated B.3 discuss and illustrate the use of these tables.

1.5.2 n AS A FUNCTION OF ES, α , AND POWER. When an investigator anticipates a certain ES, sets a significance criterion α , and then specifies the amount of power he desires, the n which is necessary to meet these specifications can be determined. This (second) type of power analysis must be at the core of any rational basis for deciding on the sample size to be used in an investigation (Cohen, 1965, pp. 97-99). For example, an investigator wishes to have power equal to .80 to detect a population r of .40 (the ES) at $\alpha_2 = .05$. By the methods described in Section 3.4 in Chapter 3, he finds that he must have $n = 46$ cases to meet these specifications. (A discussion of the basis for specifying desired power and the use of power = .80 as a convention will be found in Section 2.4 of Chapter 2.)

This major type of power analysis is discussed and illustrated in the Sections B.4 (where B indexes the chapter numbers 2-8). Each of these sections contain sample size tables (numbered B.4.A) from which, given α ,

the ES, and desired power, the n is determined. A slightly different approach to n determination is employed in Chapters 9 and 10.

1.5.3 ES AS A FUNCTION OF a , n , AND POWER. A third type of power analysis is of less general utility than the first two, but may nevertheless be quite useful in special circumstances (Cohen, 1970). Here, one finds the ES which one can expect to detect for given a , n , and with specified power. For example, an investigator may pose the question, "For a significance test of a product moment r at $a_2 = .05$ with a sample of $n = 30$, what must the population r (the ES) be if power is to be .80, i.e., what is the *detectable* ES for these specifications?" The answer, obtainable by backward interpolation (in Table 3.3.5) is that the population r must be approximately .48. Were his n equal to 46, the detectable ES would be $r = .40$.

This form of power analysis may be conventionalized for use in comparisons of research results as in literature surveys (Cohen, 1965, p. 100). One can define, as a convention, a comparative detectable effect size (CDES) as that ES detectable at $a_2 = .05$ with power = .50 for the n used in the statistical test. So defined, the CDES is an inverse measure of the sensitivity of the test, expressed in the appropriate ES unit.

This type of power analysis is not discussed in detail in the ensuing chapters. However, when readers have become familiar with the use of the tables, they will find that it can be accomplished for all of the statistical tests discussed by backward interpolation in the power tables, or when it proves more convenient, in the sample size tables.

1.5.4 a AS A FUNCTION OF n , POWER, AND ES. The last type of power analysis answers the question, "What significance level must I use to detect a given ES with specified probability (power) for a fixed given n ?" Consider an investigator whose anticipated ES is a population r of .30, who wishes power to be .75, and who as an n of 50, which she cannot increase. These specifications determine the significance criterion he must use, which can be found (by rough interpolation between subtables in Table 3.4.1) to be about $a_1 = .08$, or $a_2 = .15$.

This type of analysis is very uncommon, at least partly because of the strength of the significance criterion convention, which makes investigators loath to consider "large" values of a . We have seen that this frequently means tolerating (usually without knowing it) large values of b , i.e., low power. When power issues are brought into consideration, some circumstances may dictate unconventionally large a criteria (Cohen, 1965, p. 99ff).

This type of power analysis is not, as such, further discussed in Chapters 2-10, although it is indirectly considered in some of the examples. When the reader has become familiar with the tables, it can be accomplished for all

the statistical tests discussed in this book by interpolation between subtables of the sample size tables (B.4.A), or when more convenient, between power tables (B.3.A), within the range provided for α , i.e., α_2 : .01-.20, and α_1 : .005-.10.

In summary, four types of power analysis have been described. This book is designed primarily to facilitate two of these, the solutions for power and for sample size. It is also possible, but with less ease, to accomplish the other two, solution for ES and for α , by means of backward interpolation in the tables.

1.5.5 "PROVING" THE NULL HYPOTHESIS. Research reports in the literature are frequently flawed by conclusions that state or imply that the null hypothesis is true. For example, following the finding that the difference between two sample means is not statistically significant, instead of properly concluding from this failure to reject the null hypothesis that the data do not warrant the conclusion that the population means differ, the writer concludes, at least implicitly, that there is *no* difference. The latter conclusion is always strictly invalid, and is functionally invalid as well unless power is high. The high frequency of occurrence of this invalid interpretation can be laid squarely at the doorstep of the general neglect of attention to statistical power in the training of behavioral scientists.

What is really intended by the invalid affirmation of a null hypothesis is not that the population ES is literally zero, but rather that it is negligible, or trivial. This proposition may be validly asserted under certain circumstances. Consider the following: for a given hypothesis test, one defines a numerical value i (for *iota*) for the ES, where i is so small that it is appropriate in the context to consider it negligible (trivial, inconsequential). Power ($1 - \beta$) is then set at a high value, so that β is relatively small. When, additionally, α is specified, n can be found. Now, if the research is performed with this n and it results in nonsignificance, it is proper to conclude that the population ES is no more than i , i.e., that it is negligible; this conclusion can be offered as significant at the β level specified. In much research, "no" effect (difference, correlation) functionally means one that is negligible; "proof" by statistical induction is probabilistic. Thus, in using the same logic as that with which we reject the null hypothesis with risk equal to α , the null hypothesis can be accepted in preference to that which holds that $ES = i$ with risk equal to β . Since i is negligible, the conclusion that the population ES is not as large as i is equivalent to concluding that there is "no" (nontrivial) effect. This comes fairly close and is functionally equivalent to affirming the null hypothesis with a controlled error rate (β), which, as noted above, is what is actually intended when null hypotheses are incorrectly affirmed (Cohen, 1965, pp. 100-101; Cohen, 1970). (See Illustrative Examples 2.9, 3.5, 6.8, and 9.24.) (Also, see Fowler, 1985.)

This statistically valid basis for extracting positive conclusions from "negative findings" may not be of much practical help to most investigators. If, for example, one considers a population $r = .10$ as negligible (hence, i), and plans a test of the null hypothesis (at $\alpha_2 = .05$) for power = .95 ($\beta = .05$) to detect i , one discovers that the n required is 1294; for power = .90 ($\beta = .10$), the required $n = 1047$; and for power = .80 ($\beta = .20$), $n = 783$ (Table 3.4.1). For the much more liberal specification of $r = .20$ as i , the test (at $\alpha_2 = .05$) for power = .95 ($\beta = .05$) requires $n = 319$; for power = .90 ($\beta = .10$) requires $n = 259$, and even for power = .80 ($\beta = .20$), the required $n = 194$ (Table 3.4.1). Thus, relatively large sample sizes are necessary to establish the negligibility of an ES. But if nothing else, this procedure at least makes explicit what it takes to say or imply from a failure to reject the null hypothesis that there is no (nontrivial) correlation or difference between A and B.

1.6 SIGNIFICANCE TESTING

Although the major thrust of this work is power analysis, a simple relationship between power and significance made it relatively simple in the computation of the power tables to provide an aid to significance testing which users of this handbook may find convenient. Generally, we can define the effect size *in the sample* (ES_s) using sample statistics in the same way as we define it for the population, and a statistically significant ES_s is one which exceeds an appropriate criterion value. For most of the power tables, these criterion values for significance of the sample ES (for the given α significance criterion and n) are provided in the second column of the power tables under the symbol for the ES for that test with subscript c (for criterion), e.g., d_c for the t test on means.

1.7 PLAN OF CHAPTERS 2-10

Each of the succeeding chapters presents a different statistical test. They are similarly organized, as follows:

Section 1. The test is introduced and its uses described.

Section 2. The ES index is described and discussed in detail.

Section 3. The characteristics of the power tables and the method of their use are described and illustrated with examples.

Section 4. The characteristics of the sample size tables and the method of their use are described and illustrated with examples.

Section 5. In Chapters 2-6 and 8, the use of the power tables for significance tests is described and illustrated with examples.

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